Math 140H — Honors Calculus I: Lecture Notes

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Welcome!

These notes accompany Math 140H, Honors Calculus I, Fall 2026. Each lecture here corresponds to one class session and covers more or less the material presented in class, in the same order, so that it can serve as a compact study reference. HTML "hardcopies" of the Jupyter notebook activities from class are also included. Lecture notes will generally be available immediately **after** the corresponding class session.

State College, Fall 2026

Part I

Real Numbers

1 The Rational Numbers and the Number Line

Companion reading

This lecture corresponds to C&J §1.1a, together with the neighborhood definition from §1.1d — covered a bit early here since we will use it constantly for the rest of the semester.

Calculus is built on the real numbers, so before we can talk about limits, derivatives, or integrals we need to understand exactly what a real number *is* — and, just as importantly, why the rational numbers alone are not enough. This lecture makes that precise: we recall how the natural numbers extend to the rationals, set up the number line, and then prove that a length as ordinary as the hypotenuse of a simple right triangle cannot be a rational number at all.

1.1 From Counting to Measuring

There are two major uses for numbers: **counting** and **measuring**. The natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ are the tool for counting — abstract symbols for how many elements are in a collection, stripped of any reference to what is actually being counted. They come equipped with addition and multiplication satisfying a short list of familiar rules.

Remark 1.1. For all natural numbers a, b, c:

- $a + b = b + a$ and $ab = ba$ (commutative laws)
- $a + (b + c) = (a + b) + c$ and $a(bc) = (ab)c$ (associative laws)
- $a(b + c) = ab + ac$ (distributive law)
- $a + c = b + c \;\Rightarrow\; a = b$ (cancellation law)

These are the *rules* of arithmetic, not its *content* — everything else in this course is built on top of them, and we will not dwell on them further.

Natural numbers can be added and multiplied, but not always subtracted or divided. There is no natural number a with

$$5 + a = 2,$$

so subtraction is not always possible in $\mathbb{N}$. Extending $\mathbb{N}$ to the **integers** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ fixes this: now $5 + (-3) = 2$.

In $\mathbb{Z}$, we still cannot always divide: there is no integer $b \in \mathbb{Z}$ with

$$5 \cdot b = 2.$$

Extending $\mathbb{Z}$ to the **rational numbers** fixes this in turn.

Definition 1.1.
A **rational number** is a number that can be written as p/q where $p, q \in \mathbb{Z}$ and $q \neq 0$. Every rational number has a unique representation p/q with $q > 0$ and $\gcd(p, q) = 1$ (no common factor larger than 1); we call this representation **in lowest terms**.

Within $\mathbb{Q}$, all four rational operations — addition, subtraction, multiplication, and division (except by 0) — can be carried out freely, and they obey exactly the same laws as in Remark 1.1. In this precise sense, $\mathbb{Q}$ is the smallest extension of the natural numbers in which counting-type arithmetic becomes measuring-type arithmetic.

1.2 The Number Line

To turn numbers into geometry, fix a line L, a point O on it called the **origin**, and a second point I to the right of O; the segment OI serves as our unit scale for measuring — a "prototype meter." The direction from O to I is called **positive**.

Every point on L is now uniquely determined by two pieces of information: its **distance** from O, and its **direction** (left or right) from O. For example, the point at distance $\frac{1}{2}$ in the positive direction represents the rational number $\frac{1}{2}$; the point at distance 2 in the positive direction represents 2.

Definition 1.2.
The distance of a number x from the origin is written $|x|$ and called the **absolute value** of x:

$$|x| \;=\; \begin{cases} x & x \geq 0 \\ -x & x < 0. \end{cases}$$

Note $|x| \geq 0$ always, with equality only when $x = 0$.

Every rational point can in fact be *constructed*, not just described: to locate p/q on L with ruler and compass, construct a subdivision of the segment OI into q equal parts, then extend the part closest to O, p times. We will identify each rational number with its representation on L.

1.3 Distances and Intervals

For $x, y \in L$, the number $|x - y|$ is the **distance between** x **and** y. Observe: if $x, y \in \mathbb{Q}$, then $|x - y| \in \mathbb{Q}$ as well — distances between rational points are themselves rational.

Definition 1.3.
For $a < b$, the **interval** between a and b comes in four flavors:

- open: $(a, b) := \{x : a < x < b\}$ (both endpoints excluded)
- closed: $[a, b] := \{x : a \leq x \leq b\}$ (both endpoints included)
- half-open: $[a, b) := \{x : a \leq x < b\}$ and $(a, b] := \{x : a < x \leq b\}$

Definition 1.4.
Given a number a and $\varepsilon > 0$, the ε**-neighborhood of** a is the open interval

$$\{y : |y - a| < \varepsilon\} = (a - \varepsilon, a + \varepsilon),$$

i.e. the set of points less than ε away from a.

We will meet ε-neighborhoods constantly once we get to limits — keep this definition close by.

1.3.1 Density

Proposition 1.1.
Between any two distinct rational numbers $a < b$ there is another rational number c with $a < c < b$. Consequently, between a and b there are infinitely many rational numbers.

Proof. Let $c = \dfrac{a+b}{2}$. Since $a, b \in \mathbb{Q}$ and $\mathbb{Q}$ is closed under addition and division by 2, $c \in \mathbb{Q}$; and since $a < b$, averaging gives $a < c < b$.

Now repeat the argument on the pair a, c to get a rational number strictly between a and c, hence strictly between a and b. Doing this again and again produces infinitely many rational numbers between a and b. $\square$

We say the rationals are **dense** on the number line: no matter how closely you look, you will always find another one.

1.4 Incommensurable Lengths

Given how constructible and how dense $\mathbb{Q}$ is, a natural question arises: is there anything "left in between" the rational numbers? Are there any *gaps*? Clearly, such a gap could not itself be an interval — density rules that out, since any interval of positive length contains infinitely many rational points. If a gap exists at all, it can only be a single missing point.

Remark 1.2. The discovery that not every length is a rational multiple of a chosen unit is due to the school of Pythagoras, in the fifth or sixth century BCE. Consider a right triangle with both legs of length 1; by the Pythagorean theorem, its hypotenuse a satisfies

$$1^2 + 1^2 = a^2 = 2,$$

and a can certainly be constructed with ruler and compass. The theorem below shows this length cannot be written as p/q for integers p, q — the hypotenuse and the legs of this triangle are, in the language of the ancients, **incommensurable**: there is no common unit of which both are integer multiples.

Theorem 1.1.
There is no rational number x with $x^2 = 2$. Equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Proof. Suppose, toward a contradiction, that $\sqrt{2} = p/q$ for integers p, q with $q \neq 0$ and $\gcd(p, q) = 1$ (i.e. p/q is in lowest terms).

1. Square both sides and clear the denominator: $p^2 = 2q^2$.
2. Conclude that p^2 is even — if p were odd, p^2 would be odd, but $p^2 = 2q^2$ is even. Write $p = 2r$ for some integer r.
3. Substitute $p = 2r$ into the equation from Step 1 and simplify to get an equation for q^2 in terms of r: $(2r)^2 = 2q^2 \Rightarrow 4r^2 = 2q^2 \Rightarrow q^2 = 2r^2$.
4. Conclude that q is also even, by the same reasoning as in Step 2.
5. Steps 2 and 4 show that 2 divides both p and q. This contradicts $\gcd(p, q) = 1$, which we assumed when we wrote p/q in lowest terms.
6. Conclude: $\sqrt{2}$ is irrational — no integers p, q satisfy $\sqrt{2} = p/q$, so $\sqrt{2} \notin \mathbb{Q}$.

$\square$

Exercise 1.1. Adapt the proof of Theorem 1.1 to show that $\sqrt{3}$ is irrational. Then try to see where the argument would break if you attempted the same proof for $\sqrt{4}$ — since $\sqrt{4} = 2$ is certainly rational, something in the argument *must* fail. Where, exactly?

1.5 Summary

- The rationals $\mathbb{Q}$ satisfy all the usual laws of arithmetic (Remark 1.1) and are **dense**: between any two rationals lies another (Proposition 1.1).
- Distances $|x-y|$, intervals, and ε-neighborhoods (Definition 1.3, Definition 1.4) give us the basic vocabulary for talking about "closeness" on the number line — vocabulary we will lean on heavily once limits arrive.
- Density notwithstanding, $\mathbb{Q}$ does not exhaust the number line: Theorem 1.1 exhibits an ordinary, constructible length that is not a rational number.
- The rationals therefore leave *gaps* in the line. Lecture 2 closes these gaps rigorously, using **nested intervals** to construct the real numbers.

2 The Real Numbers via Nested Intervals

Companion reading

This lecture corresponds to C&J §1.1b–c.

Last time we found a gap: the length $\sqrt{2}$ is constructible with ruler and compass, yet Theorem 1.1 shows it is not a rational number. So what *is* $\sqrt{2}$, as a number? We are used to describing such quantities through infinite decimal expansions, e.g.

$$\pi = 3.1415926\ldots,$$

but so far this is just a string of digits — it is not clear what it *means* mathematically. The key idea of this lecture is that an infinite decimal (or any other description of an irrational number) is really a recipe for **approximating** a point by rational numbers, to any desired degree of accuracy. We make this precise using shrinking intervals with rational endpoints.

2.1 Nested Intervals

Suppose x is a point on the number line that we want to pin down using only rational numbers. Start with any closed interval $I_1 = [a_1, b_1]$ with $a_1, b_1 \in \mathbb{Q}$ and $a_1 \leq x \leq b_1$. Now choose a subinterval $I_2 = [a_2, b_2] \subseteq I_1$ that still contains x:

$$a_1 \leq a_2 \leq x \leq b_2 \leq b_1,$$

for instance by splitting I_1 in half and keeping whichever half contains x. Repeating this forever produces a chain

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots, \qquad x \in I_n \text{ for every } n.$$

For this to genuinely *pin down* x — rather than just hover near it — we need one more ingredient: the intervals must shrink all the way down to a point.

Definition 2.1.
A **nested sequence of closed intervals with rational endpoints** is a sequence $I_n = [a_n, b_n]$, $n = 1, 2, 3, \ldots$, with $a_n, b_n \in \mathbb{Q}$, such that

1. $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$, and
2. the lengths $|I_n| := b_n - a_n$ tend to 0 as $n \to \infty$: for every $\varepsilon > 0$ there is an N such that $|I_n| < \varepsilon$ for all $n \geq N$.

Proposition 2.1.
If a nested sequence of closed intervals (I_n) contains a point in every I_n, that point is unique: there is at most one x with $x \in I_n$ for all n.

Proof. Suppose x and y both lie in every I_n, and toward a contradiction suppose $x \neq y$. Let $\varepsilon_0 = |x - y| > 0$. Since $|I_n| \to 0$, there is some n with $|I_n| < \varepsilon_0$. But $x, y \in I_n$ means $|x - y| \leq |I_n| < \varepsilon_0 = |x - y|$, which is absurd. Hence $x = y$. □

So a nested sequence of intervals can "address" at most one point — and, geometrically, every point x on the line does arise this way:

Theorem 2.1.
Every real number x is the unique point contained in every interval of some nested sequence of closed intervals with rational endpoints.

Proof. Uniqueness is Proposition 2.1. For existence, pick any rational bracket $I_1 = [a_1, b_1]$ with $a_1 \leq x \leq b_1$, and repeatedly bisect: given $I_n = [a_n, b_n]$, let $m_n = \frac{a_n + b_n}{2}$ (rational, since $a_n, b_n \in \mathbb{Q}$), and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } x \leq m_n \\ [m_n, b_n] & \text{if } x > m_n. \end{cases}$$

Then $x \in I_{n+1} \subseteq I_n$ and $|I_{n+1}| = |I_n|/2$, so $|I_n| = |I_1|/2^{n-1} \to 0$. This is a nested sequence of closed intervals with rational endpoints containing x. □

In this sense, every real number can be described — completely and unambiguously — by an infinite sequence of rational numbers (the endpoints a_n, b_n).

2.2 The Nested Interval Axiom

The construction above starts from a real number x and produces a nested sequence around it. What about the converse: if we are simply *handed* a nested sequence of closed intervals with rational endpoints, must there be a point in $\bigcap_n I_n$ at all?

For **open** intervals, the answer is no: $I_n = (0, \frac{1}{n})$ satisfies $I_1 \supseteq I_2 \supseteq \cdots$ and $|I_n| = \frac{1}{n} \to 0$, yet no point lies in every I_n — any candidate $x > 0$ fails once $n > 1/x$, and $x = 0$ is excluded from every I_n by openness.

For **closed** intervals, this is far less obvious, and cannot be deduced from the algebraic properties of $\mathbb{Q}$ alone (after all, $\mathbb{Q}$ itself has a gap at $\sqrt{2}$). We take it as a basic postulate about the number line.

Axiom (Nested Interval Property)

If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ is a nested sequence of closed intervals with rational endpoints, then

$$\bigcap_{n\in\mathbb{N}} I_n \neq \emptyset,$$

i.e. there is a point x contained in every I_n.

This is a genuine **axiom**: a basic postulate we accept without proof, rather than a consequence of things we already know. Together with Proposition 2.1, it guarantees that no gaps exist on the number line — every nested sequence of rational brackets zeroes in on exactly one real number, no more and no less. This completeness of the number line is fundamental for everything that follows in this course; we will meet it again, in other guises, once we get to limits.

2.3 Decimal Expansions

Decimal expansions are simply a very concrete, systematic way of producing a nested sequence of intervals — one that subdivides by tens at every step.

Definition 2.2.
Let x be a point on the number line. Set

$$c_0 := \lfloor x \rfloor,$$

the greatest integer with $c_0 \leq x \leq c_0 + 1$, and $I_0 = [c_0, c_0 + 1]$, so $|I_0| = 1$.

Divide I_0 into ten equal parts using the points $c_0 + \frac{1}{10}, c_0 + \frac{2}{10}, \dots, c_0 + \frac{9}{10}$, and let $c_1 \in \{0, 1, \dots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} \;\leq\; x \;\leq\; c_0 + \frac{c_1}{10} + \frac{1}{10},$$

and set I_1 to be this interval, so $|I_1| = \frac{1}{10}$.

Iterating, having chosen digits $c_1, \dots, c_{n-1} \in \{0, \dots, 9\}$, subdivide I_{n-1} into ten equal parts and let $c_n \in \{0, \dots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} + \cdots + \frac{c_n}{10^n} \;\leq\; x \;\leq\; c_0 + \frac{c_1}{10} + \cdots + \frac{c_n}{10^n} + \frac{1}{10^n},$$

and let I_n be this interval, so $|I_n| = 10^{-n}$.

By construction $(I_n)_{n\geq 0}$ is exactly a nested sequence of closed intervals with rational endpoints in the sense of Definition 2.1: $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$, each contains x, and $|I_n| = 10^{-n} \to 0$. We abbreviate this whole sequence by writing

$$x = c_0 + 0.c_1c_2c_3\dots,$$

the **decimal expansion** of x — a notational shortcut for the nested intervals, not a new kind of number. (Careful when $c_0 < 0$: the digits after the point are still *added* to c_0. For instance $x = -2.75$ has $c_0 = \lfloor -2.75 \rfloor = -3$ and $x = -3 + 0.25$, not $-3 - 0.25$.)

Combining Definition 2.2 with Proposition 2.1 and the Nested Interval Axiom:

- Every real number has an (infinite) decimal expansion.
- Conversely, every such expansion — i.e. every choice of $c_0 \in \mathbb{Z}$ and digits $c_1, c_2, \dots \in \{0, \dots, 9\}$ — is a nested sequence of closed intervals with rational endpoints, so by the axiom it defines a real number.

Decimal expansions are not always unique

Some real numbers have *two* decimal expansions: for instance

$$1.000\dots = 0.999\dots.$$

Exercise 2.1. Why does $1.000\dots = 0.999\dots$? (Hint: what would $1 - 0.999\dots$ have to be, if it were positive?) More generally, can you characterize *exactly* which real numbers have two decimal expansions?

Solution. If $1 - 0.999\dots = \delta > 0$, then δ would have to be smaller than 10^{-n} for every n (since $0.\underbrace{9\cdots 9}_{n} = 1 - 10^{-n}$ already exceeds $1 - \delta$), forcing $\delta = 0$ — contradiction. So the two expansions name the same point.

In general, a real number has two decimal expansions exactly when it is a **terminating** decimal, i.e. of the form $m/10^n$ for some $m \in \mathbb{Z}$, $n \geq 0$: one expansion ends in all 0's, the other in all 9's. Every other real number has a unique decimal expansion.

2.4 Other Bases

Nothing in Definition 2.2 is special about the number 10 — we could subdivide each interval into any number $p > 1$ of equal parts instead of ten, producing digits $c_i \in \{0, 1, \dots, p-1\}$ and a **base-p (or p-ary) expansion** of x.

The case $p = 2$, the **binary expansion**, is exactly the bisection procedure from the proof of Theorem 2.1: at each step we only need to decide whether x lies in the lower or upper half of the current interval.

Example 2.1.
Let's find the first three binary digits of $x = 0.76$. Start with $I_0 = [0, 1]$, so $c_0 = 0$.

- Midpoint 0.5: since $0.76 \geq 0.5$, digit $c_1 = 1$ and $I_1 = [0.5, 1]$.
- Midpoint 0.75: since $0.76 \geq 0.75$, digit $c_2 = 1$ and $I_2 = [0.75, 1]$.
- Midpoint 0.875: since $0.76 < 0.875$, digit $c_3 = 0$ and $I_3 = [0.75, 0.875]$.

So $0.76 = 0.110\ldots$ in base 2.

Exercise 2.2. Find the first three fractional digits of the base-2 expansion of $x = 0.3$, and the first three fractional digits of the base-3 expansion of $x = 0.76$.

2.5 Summary

- A **nested sequence of closed intervals with rational endpoints** (Definition 2.1) can contain at most one point (Proposition 2.1), and every real number arises as the unique point of some such sequence (Theorem 2.1).
- The converse — that every nested sequence of closed intervals with rational endpoints *does* contain a point — is not provable from what came before; it is the **Nested Interval Axiom**, the basic postulate that guarantees there are no gaps in the number line.
- **Decimal expansions** (Definition 2.2) are the special case of base-10 subdivision; every real number has one, though numbers of the form $m/10^n$ have exactly two (Exercise 2.1).
- The same construction works in any base $p > 1$, giving p-ary expansions; base 2 is exactly the bisection algorithm behind Theorem 2.1.

3 Jupyter Activity: p-ary Expansions

Math 140H, Honors Calculus I

3.1 Before we start: how to use a Jupyter notebook

This is the first notebook of the semester, so we start a quick orientation:

- A notebook is a sequence of **cells**. This text lives in a *markdown cell* (formatted text); cells with `code` in the toolbar are *code cells* (Python).
- Click a cell and press **Shift+Enter** to run it and move to the next one. Code cells must be run in order the first time through, top to bottom — a later cell may use a function or variable that an earlier cell defined.
- If something breaks, it is often because a cell was skipped or run twice out of order. Use the menu command *Run All* (or *Restart Kernel and Run All*) to run the whole notebook cleanly from the top.
- You do **not** need to know Python already. Every function below is commented line by line — read the comments (the lines starting with `#`) as a description of the math, and the code as one particular way of writing it down.

3.1.1 Code cells

You can think of a code cell very much like a fancy calculator. Move to the next cell an run it.

```
16+(53*8.7777-128)
```

```
353.2181
```

Careful: In Python, exponentiation is `**`, not `^`.

```
2**4
```

```
16
```

```
2^4
```

```
6
```

`^` in Python is the operation called `bitwise XOR`; that explains the strange result above.

We will learn more about Jupyter notebooks (and a little bit more about Python) in the course of this semester. Today, we will simply experiment a bit with pre-defined Python function that converts rational numbers to p-ary expansions. Go ahead and run the cells below one after the other (do not forget to read the text inbetween.)

3.2 Importing Python modules

A lot functionality in Python is provided by *modules*. The following cell loads the modules we need for this notebook.

```
# We import three tools before doing anything else:
#
#   math     -- the standard library module with math.floor(y), the greatest integer <=
#               (We need floor of *exact fractions*, not of floating-point approximation
#               see the note below.)
#   Fraction -- from the 'fractions' module. A Fraction stores a rational number as an e
#               numerator/denominator pair, e.g. Fraction(1, 3) is *exactly* one third,
#               This matters a lot here: a computer's ordinary decimal numbers (called '
#               are themselves binary approximations, so 1/3 as a float is already sligh
#               and the tiny error would snowball after enough steps of the algorithm be
#               Using Fraction means every digit we compute is provably correct, not jus
#   numpy    -- imported as 'np' by convention. We use it only to store the list of digi
#               compute as a numpy array, which is a convenient container for a sequence
import math
from fractions import Fraction
import numpy as np
```

The cells below show the advantage of the `Fraction` function:

```python
1/2+1/3   # will use decimal expansions
```

```
0.8333333333333333
```

```python
Fraction(1,2)+Fraction(1,3)   # will preserve the precise fraction representation
```

```
Fraction(5, 6)
```

3.3 From a rational number to its *p*-ary expansion

We generalize the nested interval construction to base p: given x, set

$$c_0 := \lfloor x \rfloor,$$

and then repeatedly zoom in. Having peeled off $c_0, c_1, \dots, c_{n-1}$, look at what is left over,

$$r_{n-1} \; = \; x - c_0 - \frac{1}{p}c_1 - \cdots - \frac{1}{p^{n-1}}c_{n-1} \; \in [0, p^{-(n-1)}),$$

multiply it by p^{n-1} so it lands back in $[0, 1)$, and read off the next digit as the integer part of p times that:

$$c_n = \lfloor p \cdot (p^{n-1} r_{n-1}) \rfloor.$$

(Multiplying by p here means zooming in by a factor of p.)

The function below does exactly this, but it tracks the *rescaled* remainder (called `remainder` in the code, always kept in $[0, 1)$) rather than r_{n-1} itself, since that is what you actually multiply by p at each step:

1. `remainder = x - floor(x)`, the fractional part of x, in $[0, 1)$.
2. Repeat: multiply `remainder` by `p`. Its integer part is the next digit; subtract that integer part off, leaving a new `remainder` back in $[0, 1)$, and go again.

```python
def rational_to_p_ary(x, p, max_digits=50):
    """
    Compute the base-p expansion of a rational number x.

    Parameters
    ----------
    x : Fraction, int, or a string like "7/12"
        The rational number to expand. (Do NOT pass a plain float such as 0.76 -- floats
        already binary approximations, so we would be expanding the wrong number. Use
        Fraction(76, 100) instead. See the discussion above.)
    p : int
        The base, an integer > 1 (p = 2 for binary, p = 10 for the usual decimal expansi
    max_digits : int, optional (default = 50)
        how many fractional digits to compute

    Returns
    -------
    integer_part : int
        c_0, the integer part of x (i.e. floor(x)).
    digits : numpy array of int
        The fractional digits c_1, c_2, c_3, ... that were computed, in order.
    repeat_start : int or None
        If the expansion is eventually periodic, the index (0-based, into `digits`) wher
        repeating block begins. If the expansion terminates exactly (all later digits ar
        this is None.
    """
    if not isinstance(p, int) or p <= 1:
        raise ValueError("the base p must be an integer greater than 1")

    # Convert x to an exact fraction, no matter what form it was given in.
    x = Fraction(x)

    # Step 1: peel off the integer part c_0 = floor(x). What remains, x - c_0, is the
    # 'fractional part' of x and always lies in [0, 1).
    integer_part = math.floor(x)
    remainder = x - integer_part

    digits = []            # the digits c_1, c_2, ... found so far, in order
    seen_remainders = {}   # remembers, for each remainder value we've seen, which step
                            # first showed up at -- this is how we notice a repeat
    repeat_start = None
```

```
    for step in range(max_digits):
        # Zooming into the next sub-interval of
        # length 1/p means multiplying the remainder by p; the integer part of the resul
        # exactly which of the p equal pieces x fell into, i.e. the next digit.
        remainder = remainder * p
        next_digit = math.floor(remainder)
        digits.append(next_digit)

        # Subtract off the digit we just read, leaving a fresh remainder back in [0, 1),
        # ready for the next iteration.
        remainder = remainder - next_digit

    return integer_part, np.array(digits, dtype=int), repeat_start
```

The cell below defines a small helper function to print an expansion in a readable form. We deliberately write it as `c_0 + 0.c_1c_2c_3...` with an explicit `+`, matching the lecture notes' warning: if c_0 is negative, the fractional digits are still *added* to it, not subtracted, so `-3.25` written the ordinary way would be dangerously ambiguous. Writing `-3 + 0.25` instead leaves no room for that mistake.

```
def format_expansion(integer_part, digits, p, repeat_start=None):
    """
    Turn the output of rational_to_p_ary into a readable string, e.g.
        format_expansion(0, np.array([1, 1, 0]), 2, None) -> '0 + 0.110  (base 2)'
        format_expansion(0, np.array([3]), 10, 0)          -> '0 + 0.(3)  (base 10)'   #
    """
    # Turn each digit into a one-character string. In any base p <= 10 these are ordinar
    # for p > 10 we would need letters (like hexadecimal's a-f), which we don't handle h
    digit_chars = [str(int(d)) for d in digits]

    if repeat_start is None:
        # No repeating block: just list every digit we have.
        fractional_part = "".join(digit_chars) if len(digit_chars) > 0 else "0"
    else:
        # Split into the digits before the repeat (head) and the repeating block itself
        # and wrap the tail in parentheses to mark it as repeating forever.
        head = "".join(digit_chars[:repeat_start])
        tail = "".join(digit_chars[repeat_start:])
        fractional_part = f"{head}({tail})" if head else f"({tail})"

    return f"{integer_part} + 0.{fractional_part}  (base {p})"
```

3.4 Checking the examples

In the lecture notes, we found the first three *binary* digits of $x = 0.76$ by hand: $c_1 = 1,\ c_2 = 1,\ c_3 = 0$, i.e. $0.76 = 0.110\ldots$ in base 2. Let's confirm that.

```
integer_part, digits, repeat_start = rational_to_p_ary(Fraction(76, 100), 2, max_digits=
print(format_expansion(integer_part, digits, 2, repeat_start))
print("first three fractional digits:", digits[:3], " -- matches c1, c2, c3 = 1, 1, 0 fr
```

```
0 + 0.110000101000111101011100001010  (base 2)
first three fractional digits: [1 1 0]  -- matches c1, c2, c3 = 1, 1, 0 from Example 2.6
```

Now find the first three fractional digits of the base-2 expansion of $x = 0.3$, and of the base-3 expansion of $x = 0.76$. Work those out on paper first – then run the cell below to check yourself.

```
ip, d, r = rational_to_p_ary(Fraction(3, 10), 2, max_digits=20)
print("0.3 in base 2:", format_expansion(ip, d, 2, r))
print("first three digits:", d[:3])
print()
ip, d, r = rational_to_p_ary(Fraction(76, 100), 3, max_digits=20)
print("0.76 in base 3:", format_expansion(ip, d, 3, r))
print("first three digits:", d[:3])
```

```
0.3 in base 2: 0 + 0.01001100110011001100  (base 2)
first three digits: [0 1 0]

0.76 in base 3: 0 + 0.20211200100201102212  (base 3)
first three digits: [2 0 2]
```

3.5 A few more instructive examples

- $1/3$ in base 10 has *no* terminating part at all – the repeating block starts immediately.
- $1/6$ in base 10 has one non-repeating digit followed by a repeating block (matches $1/6 = 0.1\overline{6}$).
- $1/10$ terminates in base 10 (as expected, $1/10 = 0.1$ exactly) but is *purely repeating* in base 2 – this is the mathematical reason `0.1` cannot be stored exactly as an ordinary floating-point number, and is why we insisted on `Fraction` above instead of Python floats.

- Negative x: $x = -11/4 = -2.75$. Its integer part is $c_0 = \lfloor -2.75 \rfloor = -3$, and the digits describe 0.25, which gets *added* to -3 (giving back -2.75), not subtracted.

```
for label, value, base, n in [
    ("1/3", Fraction(1, 3), 10, 10),
    ("1/6", Fraction(1, 6), 10, 10),
    ("1/10", Fraction(1, 10), 10, 10),
    ("1/10", Fraction(1, 10), 2, 30),
    ("-11/4", Fraction(-11, 4), 10, 10),
]:
    ip, d, r = rational_to_p_ary(value, base, max_digits=n)
    print(f"{label:>6} = {format_expansion(ip, d, base, r)}")
```

```
   1/3 = 0 + 0.3333333333  (base 10)
   1/6 = 0 + 0.1666666666  (base 10)
  1/10 = 0 + 0.1000000000  (base 10)
  1/10 = 0 + 0.000110011001100110011001100110  (base 2)
 -11/4 = -3 + 0.2500000000  (base 10)
```

3.6 Try it yourself

Pick your own rational number x and a base $p > 1$. Use `rational_to_p_ary` to get its p-ary expansion.

```
# Replace the values below with your own choice of x (as a Fraction), p.
x = Fraction(1, 2)
p = 10


ip, d, r = rational_to_p_ary(x, p, max_digits=30)
print(f"x in base {p}:", format_expansion(ip, d, p, r))
```

```
x in base 10: 0 + 0.500000000000000000000000000000  (base 10)
```

3.7 Food for thought

Question 1: For a fixed base p, which fractions will have a p-ary expansion that will eventually be identically 0? (such as 1/2 in base 10) Try to formulate a hypothesis and test it using the cell above.

Question 2: Is it true that all p-ary expansions of rational numbers eventually become periodic? If yes, why is this the case?

Part II

Functions

4 The Concept of Function

With the number line in hand, we can start doing calculus — but calculus is not really about individual numbers. It is about *dependence*: how one quantity changes as another one does. The area of a circle depends on its radius, the height of a falling stone depends on the elapsed time, the pressure of a gas depends on its volume. The mathematical object that captures such a dependence is a **function**, and getting comfortable with the vocabulary now will pay off for the rest of the course.

4.1 Functions, Domain, and Range

Definition 4.1.
A **function** f is a rule that assigns to every element x of a set D — the **domain** of f — one single, unambiguously determined object $f(x)$, its **value** at x, lying in a set E — the **codomain**. We write

$$f\colon D \to E, \qquad x \mapsto f(x).$$

In this course the domain is almost always a subset of the number line, $D \subseteq \mathbb{R}$, and the codomain is $E = \mathbb{R}$; we then call f a **real-valued function of a real variable** and write simply $f\colon D \to \mathbb{R}$.

The word *unambiguously* is the heart of the definition: for each input $x \in D$ there must be **exactly one** output $f(x)$ — no more (that would not be a rule) and no fewer (every point of the domain must get a value). The domain is part of the data of a function: $x \mapsto x^2$ on $\mathbb{R}$ and $x \mapsto x^2$ on $[0, \infty)$ are, strictly speaking, different functions.

Remark 4.1. When a function is specified by a formula with no domain stated, we adopt the convention that its domain is the **natural domain**: the largest set of real numbers for which the formula makes sense. For $x \mapsto 1/(x-1)$ this is $\mathbb{R} \setminus \{1\}$; for $x \mapsto \sqrt{x}$ it is $[0, \infty)$. We are always free to *restrict* to a smaller domain when we want to.

Definition 4.2.
The **range** (or **image**) of $f\colon D \to \mathbb{R}$ is the set of values it actually attains:

$$\operatorname{ran}(f) \;=\; \{\, y \in \mathbb{R} : y = f(x) \text{ for some } x \in D \,\}.$$

Remark 4.2. The range need not fill up the codomain. For $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, the codomain is $\mathbb{R}$ but

$$\operatorname{ran}(f) = [0, \infty),$$

since squares are never negative and every $y \geq 0$ is $(\sqrt{y})^2$. Saying "f maps into $\mathbb{R}$" is a statement about where the values are *allowed* to live; the range records where they *actually* land.

4.2 Injective Functions

Definition 4.3.
A function $f\colon D \to \mathbb{R}$ is **injective** (or **one-to-one**) if it sends distinct inputs to distinct outputs:

$$x \neq y \quad \Rightarrow \quad f(x) \neq f(y),$$

or equivalently, taking the contrapositive, $f(x) = f(y) \quad \Rightarrow \quad x = y$.

Example 4.1.

- $f(x) = x^3$ is injective on $\mathbb{R}$: if $x^3 = y^3$ then $x = y$ (taking cube roots).
- $g(x) = x^2$ is **not** injective on $\mathbb{R}$: $g(-1) = 1 = g(1)$, yet $-1 \neq 1$. Restricted to $[0, \infty)$, however, g *is* injective.

Injectivity is exactly the property that lets us run a function backwards; we return to this when we discuss inverse functions below.

4.3 Monotone Functions

Definition 4.4.
Let $f\colon D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$. For all $x, w \in D$, we say f is

- **increasing** if $x \leq w \quad \Rightarrow \quad f(x) \leq f(w)$;
- **strictly increasing** if $x < w \quad \Rightarrow \quad f(x) < f(w)$;
- **decreasing** if $x \leq w \quad \Rightarrow \quad f(x) \geq f(w)$;
- **strictly decreasing** if $x < w \quad \Rightarrow \quad f(x) > f(w)$.

A function of any of these four types is called **monotone**.

Example 4.2.

- $f(x) = x^3$ is strictly increasing on all of $\mathbb{R}$.

- $g\colon [0,\pi] \to \mathbb{R}$, $g(x) = \cos x$, is strictly decreasing: as the angle runs from 0 to π, the x-coordinate on the unit circle drops steadily from 1 to -1.

Exercise 4.1. Show that a strictly increasing function is injective (and likewise for strictly decreasing). Is the converse true — must an injective function be monotone?

4.4 Even and Odd Functions

Definition 4.5.
Suppose the domain D is symmetric about 0, i.e. $x \in D \quad\Rightarrow\quad -x \in D$. A function $f\colon D \to \mathbb{R}$ is

- **even** if $f(-x) = f(x)$ for all $x \in D$ — its graph is symmetric about the y-axis;
- **odd** if $f(-x) = -f(x)$ for all $x \in D$ — its graph is symmetric about the origin.

Example 4.3.
x^2, $|x|$, and $\cos x$ are even; x, x^3, and $\sin x$ are odd; $x + x^2$ is neither. Most functions are neither even nor odd.

Exercise 4.2. Show that every function f on a symmetric domain can be written *uniquely* as a sum $f = f_{\mathrm{e}} + f_{\mathrm{o}}$ of an even function f_{e} and an odd function f_{o}. (Hint: what must $\frac{1}{2}\big(f(x) + f(-x)\big)$ and $\frac{1}{2}\big(f(x) - f(-x)\big)$ be?)

4.5 Building New Functions from Old

Most functions we work with are assembled from simpler ones by a few standard operations.

4.5.1 Arithmetic combinations

Definition 4.6.
Let $f, g\colon D \to \mathbb{R}$ be two functions on the same domain, and $c \in \mathbb{R}$. Then

$$f + g, \qquad c\,f, \qquad f \cdot g$$

are again functions on D, defined **pointwise**: $(f+g)(x) = f(x) + g(x)$, $(c\,f)(x) = c\,f(x)$, $(f \cdot g)(x) = f(x)\,g(x)$. The **quotient** f/g, given by $(f/g)(x) = f(x)/g(x)$, is a function on the smaller domain

$$D \setminus \{\, x \in D : g(x) = 0 \,\}.$$

4.5.2 Composition

Definition 4.7.
Let $g\colon D \to \mathbb{R}$ and $f\colon D' \to \mathbb{R}$ with $\operatorname{ran}(g) \subseteq D'$. The **composition** $f \circ g$ ("first g, then f") is

$$(f \circ g)(x) \;:=\; f(g(x)), \qquad x \in D,$$

a function with domain $\operatorname{dom}(f \circ g) = \operatorname{dom}(g)$.

Example 4.4.
With $f(x) = \sqrt{x}$ and $g(x) = 1 + x^2$:

$$(f \circ g)(x) = \sqrt{1 + x^2} \quad (\text{all } x \in \mathbb{R}), \qquad (g \circ f)(x) = 1 + x \quad (x \geq 0).$$

The order matters: $f \circ g$ and $g \circ f$ are different functions.

4.5.3 Inverse functions

Definition 4.8.
If $f\colon D \to \mathbb{R}$ is injective, then for each $y \in \operatorname{ran}(f)$ there is **exactly one** $x \in D$ with $f(x) = y$. Sending each such y to that x defines the **inverse function**

$$f^{-1}\colon \operatorname{ran}(f) \to D, \qquad f^{-1}(f(x)) = x \;\text{ for all } x \in D, \qquad f(f^{-1}(y)) = y \;\text{ for all } y \in \operatorname{ran}(f).$$

Its domain is $\operatorname{ran}(f)$ and its range is D.

> Notation trap
>
> f^{-1} denotes the **inverse function**, not the reciprocal $1/f$. For $f(x) = x^3$ we have $f^{-1}(y) = \sqrt[3]{y}$, whereas $\dfrac{1}{f(x)} = x^{-3}$.

Remark 4.3. The point (a, b) lies on the graph of f exactly when (b, a) lies on the graph of f^{-1}. So the graph of f^{-1} is the mirror image of the graph of f across the line $y = x$.

Example 4.5.
$f(x) = x^3$ is injective on $\mathbb{R}$ with $\operatorname{ran}(f) = \mathbb{R}$, and $f^{-1}(y) = \sqrt[3]{y}$. By contrast $g(x) = x^2$ on $\mathbb{R}$ has no inverse (it is not injective); restricted to $[0, \infty)$ it does, namely $g^{-1}(y) = \sqrt{y}$.

4.6 Summary

- A **function** $f\colon D \to \mathbb{R}$ (Definition 4.1) assigns to each point of its **domain** D exactly one real value; the set of values attained is its **range** (Definition 4.2), which may be a proper subset of the codomain.
- f is **injective** (Definition 4.3) if distinct inputs give distinct outputs, and **monotone**

  (1) if it consistently preserves or reverses order; strict monotonicity implies injectivity (Exercise 4.1).

- On a symmetric domain, f may be **even** or **odd** (Definition 4.5), i.e. symmetric about the y-axis or the origin.
- New functions are built from old ones by pointwise **arithmetic** (Definition 4.6), **composition** (Definition 4.7), and — when f is injective — by forming the **inverse** f^{-1} (Definition 4.8), whose graph is the reflection of that of f across $y = x$.

5 The Elementary Functions

Companion reading

This lecture corresponds to C&J §1.3.

Almost every function met in a first calculus course is built out of a short catalogue of standard functions: polynomials, rational functions, powers, exponentials, logarithms, and the trigonometric functions. These are the **elementary functions**. This lecture is a tour of the list, noting for each type its natural domain, its range, and its basic shape — information we will need constantly once we start differentiating and integrating.

5.1 Polynomials

Definition 5.1.
A **polynomial** is a function of the form

$$p(x) \;=\; a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$

with real coefficients $a_0, a_1, \ldots, a_n$. If $a_n \neq 0$, the integer n is the **degree** of p.

The lowest degrees have familiar names:

degree	form	name
0	$y = a_0$	constant
1	$y = a_0 + a_1 x$	linear
2	$y = a_0 + a_1 x + a_2 x^2$	quadratic
3	$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3$	cubic

A polynomial is a finite sum of constant multiples of the **power functions** $1, x, x^2, x^3, \ldots$, each of which is defined for every real number. Hence every polynomial has natural domain

$$\mathrm{dom}(p) = \mathbb{R}$$

(unless we deliberately restrict it).

Remark 5.1. The pure powers x^n share a few features worth remembering. Every graph passes through $(0,0)$ and $(1,1)$. On the interval $[0,1]$ a higher power lies *below* a lower one ($x^3 \le x^2 \le x$), while on $[1,\infty)$ the order reverses. The power x^n is even when n is even and odd when n is odd; for odd n it is strictly increasing on $\mathbb{R}$ (hence injective), while for even n it is not injective on $\mathbb{R}$.

5.2 Rational Functions

Definition 5.2.
A **rational function** is a quotient of two polynomials,

$$r(x) \;=\; \frac{p(x)}{q(x)}, \qquad q \not\equiv 0.$$

Its natural domain is $\mathbb{R} \setminus Z$, where

$$Z = \{\, x \in \mathbb{R} : q(x) = 0 \,\}$$

is the (finite) set of zeros of the denominator.

Example 5.1.

- $\dfrac{x}{x^2+1}$ is a rational function. Since $x^2+1>0$ for every real x, the set Z is empty and the domain is all of $\mathbb{R}$.
- $\dfrac{\sqrt{x}}{x^3-x}$ is **not** a rational function: $\sqrt{x}$ is not a polynomial. (It is, however, still an elementary function, as it is the arithmetic combination of two elementary functions - see next section)

5.3 Algebraic and Power Functions

Definition 5.3.
A **power function** is $f(x) = x^\alpha$ with a fixed real exponent $\alpha \in \mathbb{R}$. For $x > 0$ this is defined for every α (via $x^\alpha = \exp(\alpha \ln x)$, once we have exp and ln). For an **integer** exponent the domain extends to

$$\mathrm{dom}(x^\alpha) = \begin{cases} \mathbb{R} & \alpha \ge 0, \\ \mathbb{R} \setminus \{0\} & \alpha < 0. \end{cases}$$

Remark 5.2. For a non-integer exponent one usually keeps the domain $[0, \infty)$ (or $(0, \infty)$ if $\alpha < 0$). The roots $x^{1/2} = \sqrt{x}$, $x^{1/3} = \sqrt[3]{x}, \ldots$ are examples; odd roots such as $\sqrt[3]{x}$ can in fact be extended to all of $\mathbb{R}$. Power functions with rational exponent, and more generally functions obtained from polynomials by the four arithmetic operations together with root extraction, are called **algebraic functions**. The exponential and logarithm below are *not* algebraic — they are **transcendental**.

5.4 Exponential Functions

Definition 5.4.
Fix a **base** $a > 0$. The **exponential function** with base a,

$$\exp_a : \mathbb{R} \to \mathbb{R}, \qquad \exp_a(x) = a^x,$$

is defined for every real x. For $a \neq 1$ it takes only positive values, and every positive value is attained:

$$\mathrm{dom}(\exp_a) = \mathbb{R}, \qquad \mathrm{ran}(\exp_a) = (0, \infty) = \mathbb{R}^{>0}.$$

For $a > 1$ the function a^x is strictly increasing; for $0 < a < 1$ it is strictly decreasing; for $a = 1$ it is the constant 1. In all cases $a^0 = 1$ and $a^{x+y} = a^x a^y$.

5.5 Logarithms

Definition 5.5.
Fix a base $a > 0$ with $a \neq 1$. Then $\exp_a$ is injective from $\mathbb{R}$ onto $(0, \infty)$, and its inverse function (in the sense of Definition 4.8) is the **logarithm to base** a,

$$\log_a : (0, \infty) \to \mathbb{R}, \qquad \log_a(a^x) = x, \qquad a^{\log_a y} = y.$$

Thus

$$\mathrm{dom}(\log_a) = (0, \infty), \qquad \mathrm{ran}(\log_a) = \mathbb{R}.$$

Because it is the inverse of a^x, the graph of $\log_a$ is the reflection of the graph of a^x across the line $y = x$ (Remark 4.3). It converts products to sums: $\log_a(xy) = \log_a x + \log_a y$.

5.6 Trigonometric Functions

Let x be a real number, thought of as an angle measured in **radians**. Rotating the point $(1, 0)$ counter-clockwise through the angle x about the origin lands on a point of the unit circle whose coordinates are, by definition, $(\cos x, \sin x)$.

Both sin and cos are defined for all $x \in \mathbb{R}$, take values in $[-1, 1]$, and repeat after a full turn.

Definition 5.6.
A function f is **periodic with period** $P > 0$ if $f(x + P) = f(x)$ for every x in its domain.

Thus sin and cos are periodic with period 2π. The **tangent**

$$\tan x = \frac{\sin x}{\cos x}$$

is defined wherever $\cos x \neq 0$, i.e. for $x \neq \frac{\pi}{2} + k\pi$ with $k \in \mathbb{Z}$, and is periodic with period π.

Remark 5.3. A general **sinusoid** has the form

$$f(x) = A\,\sin(bx) + C,$$

with **amplitude** $|A|$ (half the vertical spread of the graph), **period** $\dfrac{2\pi}{b}$, and **vertical offset** C (the horizontal line the graph oscillates around).

Exercise 5.1. The function $f(x) = 3\sin(2x) + 1$ is a sinusoid. Find its amplitude, period, and range, and the smallest $x > 0$ at which it attains its maximum.

5.7 Summary

- **Polynomials** (Definition 5.1) are finite combinations of the powers $1, x, x^2, ...$; they are defined on all of $\mathbb{R}$.
- **Rational functions** (Definition 5.2) are quotients of polynomials, defined off the finite zero set of the denominator.
- **Power functions** x^α (Definition 5.3) have domain $\mathbb{R}$ or $\mathbb{R} \setminus \{0\}$ for integer exponents, and are typically taken on $[0, \infty)$ otherwise; together with roots they form the **algebraic** functions.
- **Exponentials** a^x (Definition 5.4) have domain $\mathbb{R}$ and range $(0, \infty)$; their inverses, the **logarithms** $\log_a$ (Definition 5.5), have domain $(0, \infty)$ and range $\mathbb{R}$.

- The **trigonometric functions** $\sin, \cos$ are 2π-periodic (Definition 5.6) with range $[-1, 1]$; $\tan = \sin / \cos$ is π-periodic and undefined where $\cos$ vanishes.

6 Continuity

Companion reading

This lecture corresponds to C&J §1.2.d.

Intuitively, a function is continuous if a small change in the independent variable produces only a small change in the dependent variable — the graph has no jumps, gaps, or blow-ups. Before making this precise, it helps to see why the intuitive idea is not yet a *definition*: it uses the word "small" twice without saying how small, or how the two smallnesses are related.

Example 6.1.
Let

$$f(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0. \end{cases}$$

At $x_0 = 0$, an arbitrarily small change in x — moving from $x = 0$ to some $x < 0$ — causes f to jump from 1 to 0, a change of a full unit. No matter how small a step we take to the left of 0, the output change stays large. We call this a **jump discontinuity** at $x_0 = 0$.

Q: How do we make "small change forces small change" precise? The idea is to formalize "small" using neighborhoods.

Definition 6.1.
Given $a \in \mathbb{R}$ and $\varepsilon > 0$, the ε**-neighborhood** of a is the interval

$$(a - \varepsilon,\, a + \varepsilon).$$

Think of ε as a *tolerance*: saying "x is within tolerance of a" means $x \in (a-\varepsilon, a+\varepsilon)$, i.e. $|x - a| < \varepsilon$. "Small change" becomes "ε small."

Fix a point x_0. We want: the distance between $f(x)$ and $f(x_0)$ stays within tolerance ε, i.e.

$$|f(x) - f(x_0)| < \varepsilon,$$

whenever x is *sufficiently close* to x_0 — say, within some tolerance δ of x_0. The key point is that this has to work for **any** $\varepsilon > 0$, no matter how small; but we are allowed to choose δ *in response to* ε — a smaller tolerance on the output may well force a smaller tolerance on the input.

6.1 The ε–δ Definition

Definition 6.2.
Let $f\colon D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$, and let $x_0 \in D$. We say f is **continuous at** x_0 if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x \in D \text{ and } |x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

Example 6.2.
Take f as in Example 6.1 and $x_0 = 0$. Pick $\varepsilon = \frac{1}{2}$. No matter how small we choose $\delta > 0$, there is always some $x < 0$ with $|x - x_0| < \delta$ (i.e. x is δ-close to $x_0 = 0$), and for any such x,

$$|f(x) - f(x_0)| = |0 - 1| = 1 > \tfrac{1}{2}.$$

So no δ works for this ε, and f is **not** continuous at $x_0 = 0$.

Remark 6.1. Continuity is a property of f **at a point**. The function f above is continuous at every $x_0 \neq 0$ (Exercise 6.1) and discontinuous only at $x_0 = 0$ — a function can be continuous at some points of its domain and discontinuous at others.

Exercise 6.1. Show that f from Example 6.1 is continuous at every $x_0 \neq 0$.

6.1.1 The ε–δ game

Remark 6.2. It helps to think of the definition as a game between an opponent and you. The opponent challenges you with some $\varepsilon > 0$; you must exhibit a matching $\delta > 0$ so that

$$|x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

f is continuous at x_0 exactly when you can win this game **no matter which ε the opponent picks**.

Example 6.3.
$g(x) = 3x + 2$ is continuous at every $x_0 \in \mathbb{R}$.

Proof. Let $\varepsilon > 0$ and $x_0 \in \mathbb{R}$ be arbitrary. We want $|g(x) - g(x_0)| < \varepsilon$. Compute

$$|g(x) - g(x_0)| = |(3x + 2) - (3x_0 + 2)| = |3x - 3x_0| = 3|x - x_0|.$$

So if we let $\delta = \varepsilon/3$, then

$$|x - x_0| < \delta \quad \Rightarrow \quad |g(x) - g(x_0)| = 3|x - x_0| < 3\delta = \varepsilon.$$

$\square$

Exercise 6.2. Show, directly from Definition 6.2, that $h(x) = \frac{1}{2}x - 1$ is continuous at every $x_0 \in \mathbb{R}$. (What choice of δ in terms of ε makes the argument work?)

A Jupyter demonstration accompanying this lecture lets you play the ε–δ game interactively: pick x_0 and a δ, and watch the resulting range of f-values shrink (or fail to shrink) as $\delta \to 0$.

6.2 Types of Discontinuities

If f fails to be continuous at x_0, the failure typically has one of three flavors.

Remark 6.3. **Jump discontinuities.** The values of f approach one number from the left of x_0 and a different number from the right, as in Example 6.1. We will make this precise next week with *one-sided limits*: a jump discontinuity is exactly a point where the one-sided limits exist but disagree.

Remark 6.4. **Unbounded (infinite) discontinuities.** Consider

$$f(x) = \begin{cases} 1/x^2 & x \neq 0 \\ 0 & x = 0. \end{cases}$$

As $x \to 0$, $f(x) \to \infty$, so $|f(x) - f(0)|$ is **unbounded** on every neighborhood of 0 — no ε-tolerance, however large, can be met near $x_0 = 0$ by any δ.

Remark 6.5. **Oscillation discontinuities.** Consider

$$g(x) = \begin{cases} \sin(1/x) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Near $x_0 = 0$, g oscillates wildly: arbitrarily close to 0 we can always find points where g equals 1 and points where g equals -1. Indeed, for any $n \in \mathbb{N}$, taking

$$x_1 = \frac{1}{2\pi n + \pi/2} \qquad \text{gives} \qquad \sin(1/x_1) = 1,$$

and similarly $x_{-1} = \dfrac{1}{2\pi n + 3\pi/2}$ gives $\sin(1/x_{-1}) = -1$; both $x_1, x_{-1} \to 0$ as $n \to \infty$. So on every neighborhood of 0, no matter how small, g takes values ε apart for ε close to 2 — g cannot be made continuous at 0 by any choice of $g(0)$.

Example 6.4.
Contrast this with

$$x \sin(1/x) \quad (x \neq 0), \qquad \text{value 0 at } x = 0,$$

which **is** continuous at $x_0 = 0$: although $\sin(1/x)$ still oscillates without settling down, it is squeezed between -1 and 1, so $|x \sin(1/x) - 0| \leq |x| \to 0$ as $x \to 0$. Oscillation alone does not cause a discontinuity — it must fail to be *damped out* as $x \to x_0$.

6.3 Summary

- The intuitive idea "small change in, small change out" is formalized using **ε-neighborhoods** (Definition 6.1): a tolerance interval $(a - \varepsilon, a + \varepsilon)$ around a point.
- f is **continuous at** x_0 (Definition 6.2) if for every tolerance $\varepsilon > 0$ on the output, there is a matching tolerance $\delta > 0$ on the input, so that x within δ of x_0 forces $f(x)$ within ε of $f(x_0)$ — think of it as a game (Remark 6.2) you must be able to win against any ε.
- Continuity is a property of f **at a point** (Remark 6.1); a function can be continuous at some points and not others.
- Failures of continuity come in three standard flavors: **jump** discontinuities (Remark 6.3), where one-sided behavior disagrees; **unbounded** discontinuities (Remark 6.4), where f blows up; and **oscillation** discontinuities (Remark 6.5), where f never settles down near x_0 — unless the oscillation is damped, as in $x \sin(1/x)$ (Example 6.4), which remains continuous.

7 Activity: Properties of Functions

About this activity

An in-class, pencil-and-paper worksheet. It revisits the properties of functions built up in the previous chapters — *injective*, *monotone*, *even/odd*, and, above all, *continuous* — by working them out in full for two specific functions. Companion reading: C&J §1.2.d.

7.1 Background

We have met several properties a function can have: **injective**, **monotone**, **even or odd**, and — most important — **continuous**. In this activity we investigate all of them for the two functions

$$f(x) = \begin{cases} \dfrac{x}{|x|} & x \neq 0, \\ 0 & x = 0, \end{cases} \qquad g(x) = \begin{cases} \dfrac{1}{x^2} & x \neq 0, \\ 0 & x = 0. \end{cases}$$

7.2 Step 1: Draw the graphs

Sketch the graph of each function on its own set of axes. Plot a few sample points first (e.g. $x = \pm\frac{1}{2},\ \pm 1,\ \pm 2,\ \pm 3$), then connect them.

7.3 Step 2: Make an educated guess

From your graphs, try to answer the following.

1. Is the function injective?
2. Is it monotone? If so, is it *strictly* monotone? If not, can you find intervals — as large as possible — on which it is (strictly) monotone?
3. Is it even, odd, or neither?

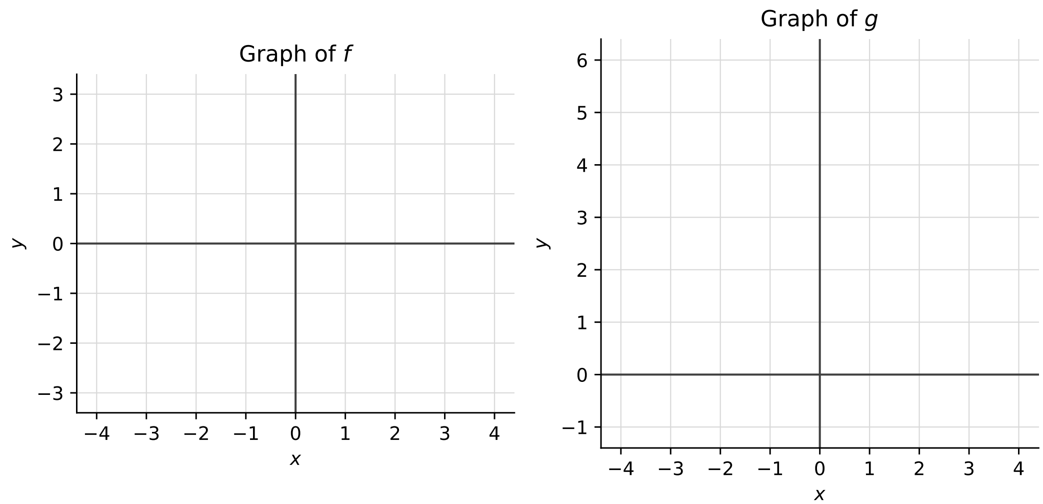

Figure 7.1: Blank axes for sketching f (left) and g (right).

4. Is it continuous everywhere? If not, at which x is it discontinuous?

Record short answers in the table; you will verify each one in the steps that follow.

Property	f	g
Injective? (yes / no)		
Monotone? Strictly? On which interval(s)?		
Even, odd, or neither?		
Continuous everywhere? If not, at which x?		

7.4 Verification: injectivity

Neither f nor g is injective. To prove this, it is enough to exhibit two inputs $x \neq y$ with the same output.

Exercise 7.1. Find explicit values:

- for f: $x =$ ____ and $y =$ ____ with $f(x) = f(y)$;
- for g: $x =$ ____ and $y =$ ____ with $g(x) = g(y)$.

7.5 Verification: even / odd

Exercise 7.2. For whichever symmetry you guessed, verify it directly from the definition of the function: compute $f(-x)$ and check that it equals $-f(x)$ (odd) or $f(x)$ (even). Then do the same for g.

7.6 Verification: monotonicity

Exercise 7.3. Verify your monotonicity guess on a suitable interval, directly from the definition. For instance, to show a function is increasing on an interval, assume $x \le y$ in that interval and deduce $f(x) \le f(y)$ (use strict inequalities for *strictly* increasing); argue similarly for decreasing.

7.7 Continuity: the ε–δ game

Both f and g are continuous at $x_0 = 1$. Play the ε–δ game from Definition 6.2.

Exercise 7.4.

1. Take $\varepsilon = 1$. Find a number $\delta > 0$ such that

$$|x - 1| < \delta \quad \Rightarrow \quad |f(x) - f(1)| < 1$$

   (recall $f(1) = 1$). Then find such a δ for g.
2. Repeat with a smaller tolerance, $\varepsilon = \frac{1}{10}$. Does the argument really change?
3. Can you produce a δ that works for an **arbitrary** $\varepsilon > 0$?

7.8 Discontinuities

Both functions are discontinuous at $x_0 = 0$. Here it is *you* who loses the game.

Exercise 7.5. Find an $\varepsilon > 0$ for which **no** δ works — that is, however close x is taken to 0, there is always some such x with

$$|f(x) - f(0)| \ge \varepsilon.$$

Give such an ε for f, and one for g.